

Labor Input Responses to Ownership Changes: Evidence from U.S. Nursing Homes

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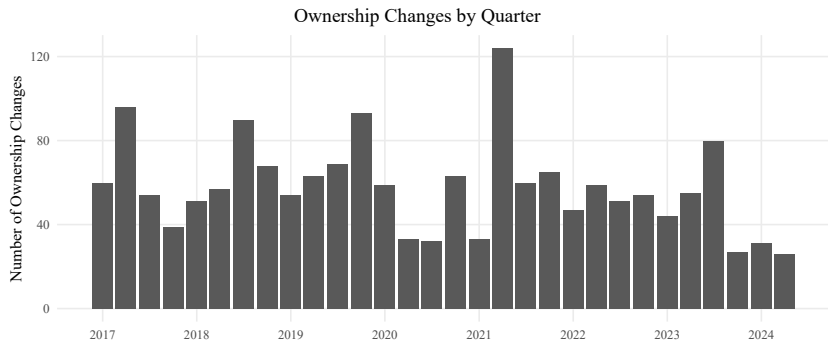
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- Instead, effects operate through **internal production decisions**.
- This type of change is understudied in the literature, and we examine it in **nursing homes**.

Ownership Changes Over Time (Quarterly)



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- Causal evidence shows that increasing nurse staffing improves measured nursing home quality ([Lin, 2014](#)).
- If ownership changes shift staffing decisions, they can translate into meaningful differences in care quality.

Research Question

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 - ▶ Do responses differ pre vs post-pandemic?

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- Ownership change can affect both **levels** and **mix** which sets up cost-quality trade off.

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- **Both margins raise costs** because nursing labor is a major operating expense.
- **But short-run revenues are relatively fixed**, so staffing choices map into a **cost–quality tradeoff**.
- **Implication:** an ownership change that shifts objectives or constraints can move a facility along this tradeoff. ([Park and Werner, 2011](#); [Bowblis and Brunt, 2025](#))

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 - ▶ Standardized staffing model shifts RN/LPN/CNA composition; total hours may be similar

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- **Facility-month panel, 48 contiguous states, Jan 2017 – Jun 2024**

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- Therefore, we have to use **NHC ownership shares** to verify and date whether control actually changes.

Ownership Level Challenge

- **NHC**
- We identify ownership changes at the **highest level of ownership present**

Ownership Level	Type	Owner Name	Share
<i>Direct ownership</i>			
Organization	LLC	BREDLEGS HOLDINGS LLC	25%
Organization	LLC	ROCK HILL ACQUISITION III LLC	75%
<i>Indirect ownership</i>			
Individual	Person	LALLY, THOMAS	25%
Individual	Person	LITT, ALAN	25%
Individual	Person	LITT, JONATHAN	25%
Individual	Person	STURM, JOSHUA	25%

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- **Example from the data:**

- ▶ *Before:*

- ★ ESTES JAMES (84%)
- ★ BURCHFIELD JOHN (10%)
- ★ LEE CLAUDE (6%)

- ▶ *After:*

- ★ JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
- ★ JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)

- ▶ We do **not** treat this as a meaningful ownership change

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- **Event month:** first month NHC shows a majority-control transfer.

Cross-tab of NHC & HCRIS

Cross-tab of Ownership Changes

Count of changes in NHC	2+	1,137 (8.5%)	1,377 (10.3%)	535 (4.0%)
	1	1,874 (14.0%)	1,589 (11.9%)	101 (0.8%)
	0	6,217 (46.6%)	499 (3.7%)	15 (0.1%)
		0	1	2+
		Count of changes in HCRIS		

Dependent Variables: Nurse Staffing (PBJ) and Sample

- We study four staffing measures:
 - ▶ RN, LPN, CNA, and total HPRD
- Outcome is **hours per resident day (HPRD)**: staffing intensity standardized by resident census

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- **Final sample:** 7,806 facilities; 632,231 facility-months; 1,589 treated ($\approx 20\%$)

Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD around ownership transitions
[Goodman-Bacon, 2021](#)
- **Variation:** within-facility staffing changes relative to facilities not yet treated
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$$Y_{it} = \sum_{\tau=-24}^{24} \beta_{\tau} \mathbf{1}\{event_time_{it} = \tau\} \cdot Treat_i + X_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}.$$

- ▶ Y_{it} : staffing outcome in **HPRD**
- ▶ β_{τ} : effect on Y_{it} at event time τ
- ▶ X_{it} controls (beds, occupancy, payer mix, ownership type, chain, acuity)
- ▶ α_i facility FE; λ_t month FE
- ▶ SEs clustered at the facility and month level

Empirical Strategy: TWFE Difference-in-Differences

- **Goal:** summarize the **average** staffing change after an ownership transition
- **TWFE DiD regression:**

$$Y_{it} = \beta Post_{it} + X_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

- ▶ $Post_{it} = 1$ in all months after a change
- ▶ β : **average post-change effect** on staffing Y_{it}
- ▶ Same fixed effects and controls as the event-study specification
- ▶ SEs clustered at the facility and month level

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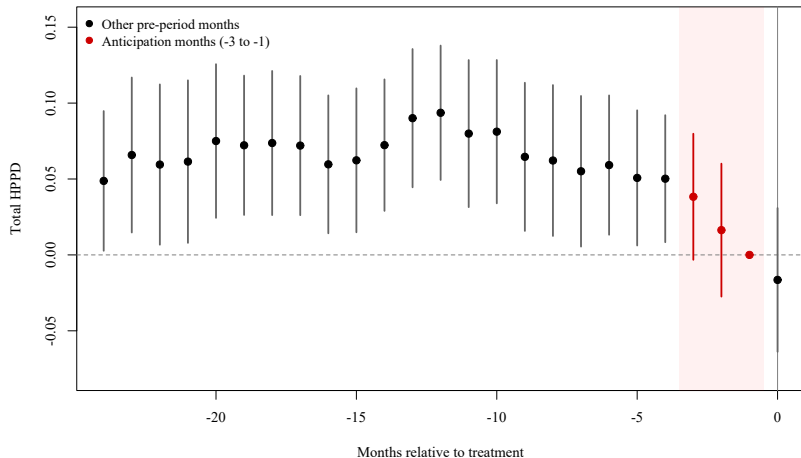
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- **Implication:** months just before $\tau = 0$ may violate parallel trends.

Evidence of violation in the Pre-Period



Main Specification: Donut Event Study + Pre-trend Test

- **Donut event study:** mitigate this influence by excluding months just before the recorded transition.
- **Implementation**
 - ▶ Exclude $\tau \in \{-3, -2, -1\}$
 - ▶ Use $\tau = -4$ as the reference period

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- **Implementation**
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 - ▶ Use $\tau = -4$ as the reference period
- **Parallel Trends:** joint Wald test of pre-period leads

$$H_0 : \beta_\tau = 0 \quad \forall \tau < 0$$

- **Takeaway:** full pre-window shows evidence of pre-trends; the donut specification is more consistent with parallel trends.

Parallel Trends Tests (Joint Wald) — Total Staffing

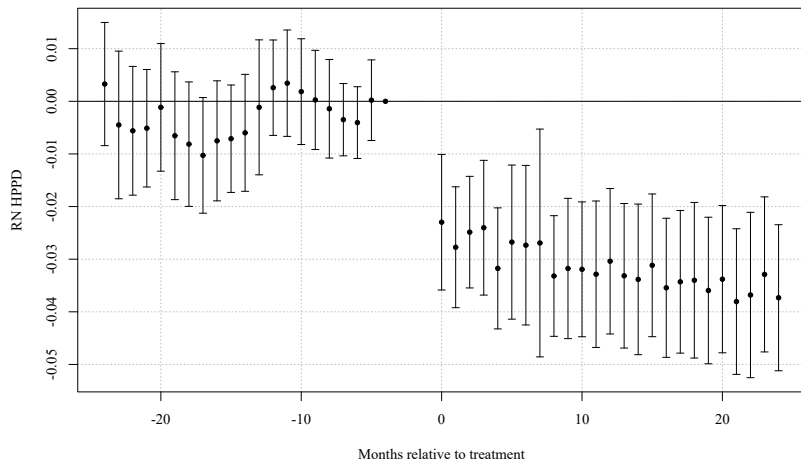
Table: Pre-trend Wald Test p-values (Total Staffing)

	Total (Levels)	Total (Logs)
2 Year Full Pre-Window	0.0006	0.0012
2 Year Window with Donut	0.2046	0.3268
1 Year Window with Donut	0.6387	0.6102

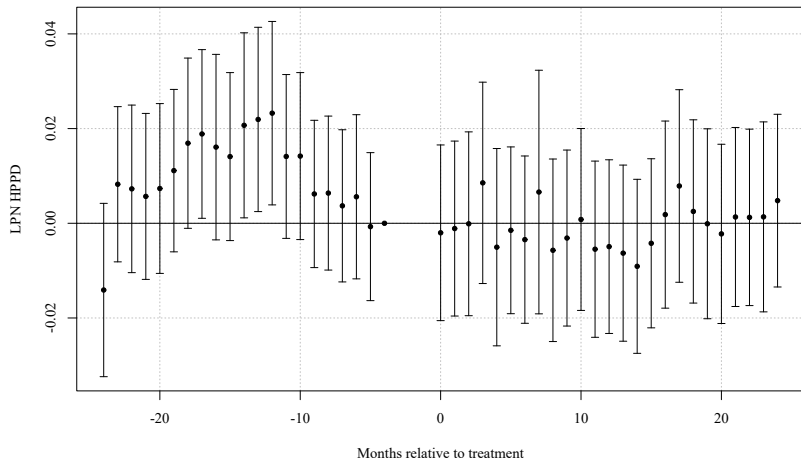
Notes: Joint Wald χ^2 test of $H_0 : \beta_\tau = 0$ for all pre-treatment coefficients. Cells report p-value.

Donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

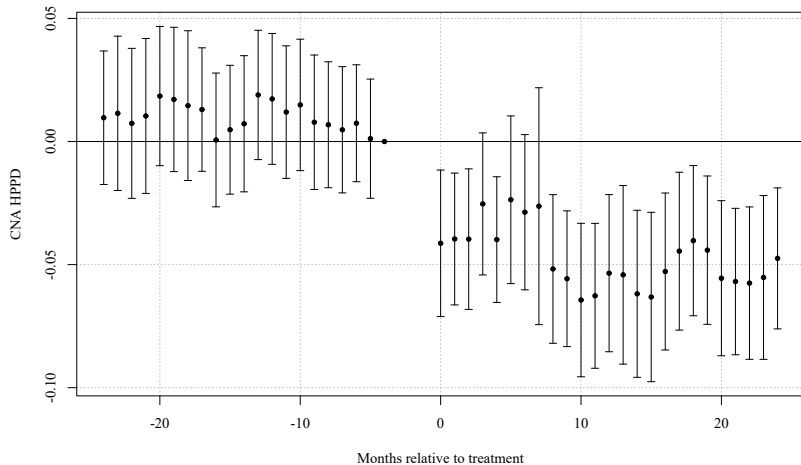
Event Study Results (RN HPRD)



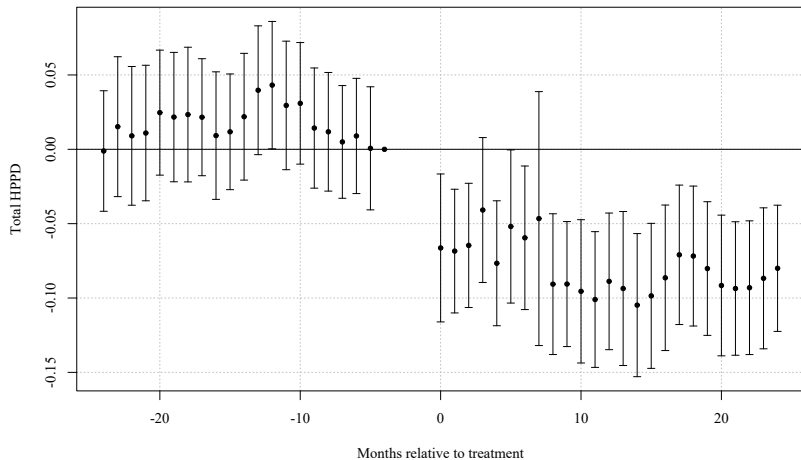
Event Study Results (LPN HPRD)



Event Study Results (CNA HPRD)



Event Study Results (Total Nursing HPRD)



TWFE DiD — Baseline (RN)

	RN	LPN	CNA	Total
HPRD	−0.032*** (0.005)	0.000 (0.006)	−0.054*** (0.009)	−0.086*** (0.013)
Log(HPRD)	−0.092*** (0.016)	0.003 (0.009)	−0.028*** (0.005)	−0.026*** (0.004)

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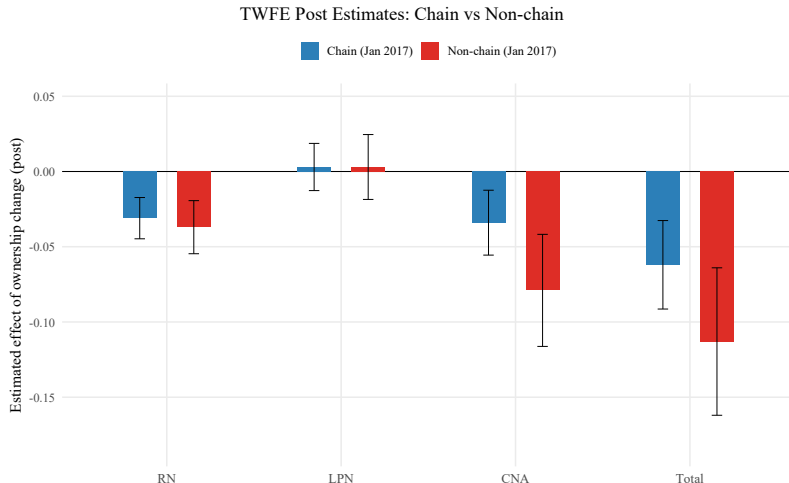
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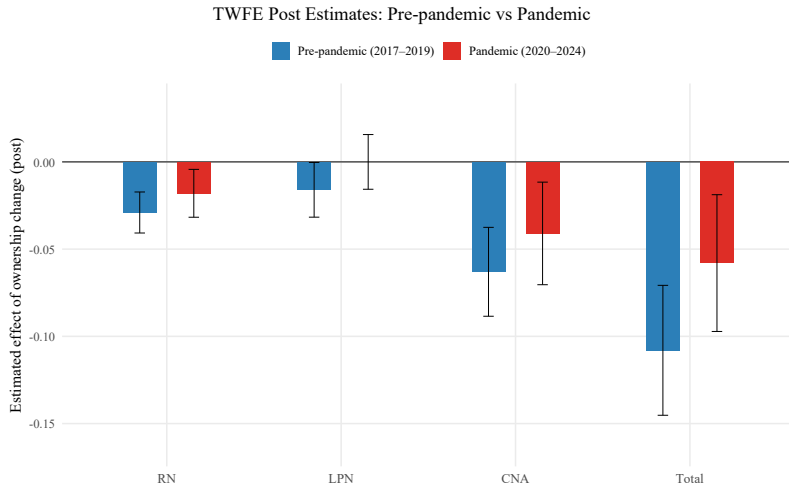
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Heterogeneity: Chain vs Non-Chain Facilities



Staffing declines are larger for non-chain facilities, nearly double for CNA and Total.

Heterogeneity: Pre vs Post-Pandemic



Large pre-pandemic declines, and post-pandemic effects are almost cut in half.

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- Takeaway: ownership changes affect care via **internal production decisions**, even without market- structure change

Thank you

Questions?

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